Ramesh Mandamanedi

AI Engineer | LLM Agents | Backend & Distributed Systems | ML Infrastructure

San Jose, CA | +1 510-598-0960 | mandamanediramesh26@gmail.com | github.com/Ramesh-codex

SUMMARY

Computer Science master's graduate building Python-based LLM agents, search infrastructure, and streaming ML systems. Developed a 15-bug agent evaluation harness and a 100K-document BM25 search benchmark; shipped two public Docker/FastAPI search applications with GitHub Actions and live HTTPS verification.

TECHNICAL SKILLS

Languages: Python, Go, C, C++, JavaScript, SQL, Bash
AI / ML: LangGraph, Anthropic API (Claude), agent evaluation, tool calling, Pydantic, PyTorch, Transformers, FinBERT
Backend / Data: FastAPI, REST APIs, Redis Streams, Redis, PostgreSQL, SQLite, distributed search, BM25
Infrastructure: Docker, Kubernetes, AWS, Terraform, GitHub Actions, GitLab CI/CD, Prometheus, Grafana, pytest

SELECTED ENGINEERING PROJECTS

Agent Bugfixer: Multi-Agent LLM Bug Fixer | Python, LangGraph, Claude, Docker | GitHub

- Built Researcher → Coder → Reviewer agent workflow with Pydantic-validated handoffs, enforced per-role tool permissions, SQLite checkpoints, and human approval before opening draft GitHub pull requests.
- Built a tamper-resistant, 15-bug Docker evaluation harness (no network, 512 MB RAM, 1 CPU); over 45 runs per setup, measured 91.1% single-agent vs. 88.9% multi-agent fix rates at ~2× estimated cost.
- Traced reviewer-induced test tampering and revised review rules; achieved 40/40 correct approvals and 5/5 failure escalations. Planted .env-secret prompt injections received zero compliance in 135 runs.

Distributed Search Engine + Developer Docs Search | Python, FastAPI, Docker, BM25 | GitHub

- Implemented tokenizer, stemmer, inverted index, and heap-based BM25 top-K without IR libraries; original 100K-Wikipedia-article benchmark: 427 docs/sec, 1.15 GB memory, p50 1.53 ms / p99 107 ms query latency.
- Deployed two public HTTPS demos: a three-shard HTTP scatter–gather search service (48-doc glossary) and documentation search across 188 excerpts from 20 official FastAPI, HTTPX, and Docker pages; verified Docker, endpoints, and all three shards in CI.
- Corrected misleading in-process memory measurements by isolating RSS sampling in subprocesses; hosted demo workloads and benchmarks use different datasets and hardware.

Live demos: Distributed Search | Developer Docs Search | Live API verification

Real-Time Stock Sentiment Pipeline | Python, FastAPI, Redis Streams, FinBERT | GitHub

- Built RSS ingestion with Redis SADD deduplication, at-least-once Redis Streams consumer groups, FinBERT inference, and FastAPI per-ticker sentiment and Prometheus metrics.
- Measured p50 46 ms / p95 65 ms CPU inference and 8–10 headlines/sec sustained; reduced Docker image 86% (11.7 → 1.6 GB) and build time from 22 to 3 minutes.

Infrastructure Automation for AI Workloads | Python, Redis, Prometheus | GitHub

- Built a Python scheduler with Redis queues and three workers, instrumented with Prometheus/Grafana; diagnosed retry storms and reduced 15-minute task failure rate ~52% (24.7% → 13.0%).
- Introduced I/O concurrency limits and exponential backoff; identified instrumentation error that counted semaphore-blocked tasks as active I/O.

EDUCATION

Master of Science, Computer Science | San Francisco Bay University, Fremont, CA | Aug 2024 – May 2026
Bachelor of Technology, Computer Science & Engineering (AI) | Parul University, India | Aug 2020 – May 2024